

Lab 3: Architectural Justification

Clinical Trial Subject Recruitment Engine

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Architecture Selection:

We chose Microservices Architecture for the Clinical Trial Subject Recruitment Engine.

Reason 1 – Independent Scaling for a Compute-Heavy Service

The Eligibility Evaluation Service must satisfy NF-02 (returning eligibility results within a strict latency target for 95% of requests, even under peak load), while Consent Management and Visit Compliance are comparatively low-throughput, form-driven workflows. Microservices let the Eligibility Evaluation Service be scaled horizontally on its own adding instances only where the compute-heavy rule evaluation happens without over provisioning the rest of the system, which a single monolithic layer could not do as cheaply.

Reason 2 – Fault Isolation Across Independent Trial Workflows

Candidate intake, consent tracking, visit scheduling, and researcher reporting are used by different actors (Trial Candidate vs. Clinical Researcher) at different times and cadences. Deploying them as independent services means a defect or outage in one workflow does not take down candidate screening or consent recording an important property for a system supporting an active clinical trial where candidate intake cannot be allowed to stall.

Security Advantage

Microservices allow security-sensitive audit logging to be centralised into its own dedicated Audit & Security Logging Service that every other service calls through a well-defined *LogEvent* interface. Because this is the only path by which state-changing events are recorded, we get a single, consistent enforcement point for the immutable, timestamped audit trail required by NF-01 rather than relying on every service to implement logging correctly on its own. Service-level isolation also limits the blast radius if any one component were compromised.

Performance Benefit

Because the Eligibility Evaluation Service is decoupled from Consent Management, Visit Compliance, and the Researcher Dashboard, it can be scaled and optimised independently to meet the NF-02 target of returning results within the required latency window for 95% of requests at peak load, without contention from unrelated, lower-priority workflows competing for the same compute resources.

System Components & Interface Specifications

Components List

Sr.No	Component	Layer	Maps to Requirements
1	Candidate Portal Service	Presentation	FR-01 (collect & validate candidate info)
2	Researcher Dashboard Service	Presentation	FR-05 (researcher review of eligibility/consent/visits)
3	Eligibility Evaluation Service	Business	FR-02, NF-02 (inclusion/exclusion screening, latency target)
4	Consent Management Service	Business	FR-03 (informed consent tracking)
5	Visit Compliance Service	Business	FR-04 (scheduled visit tracking)
6	Audit & Security Logging Service	Cross-cutting	NF-01 (immutable audit trail)
7	Trial Data Store	Data	Persistence for candidates, consent, visits

Interfaces (6)

#	Interface	Provided by	Required by	Purpose
1	CandidateSubmission	Eligibility Evaluation Service	Candidate Portal Service	Candidate submits screening form data
2	StatusQuery	Eligibility Evaluation, Consent Management, Visit Compliance Services	Researcher Dashboard Service	Read-only status lookups for researchers
3	EligibilityDataAccess	Trial Data Store	Eligibility Evaluation Service	Persist/retrieve eligibility records
4	ConsentDataAccess	Trial Data Store	Consent Management Service	Persist/retrieve consent records
5	VisitDataAccess	Trial Data Store	Visit Compliance Service	Persist/retrieve visit records
6	LogEvent	Audit & Security Logging Service	Eligibility Evaluation, Consent Management, Visit Compliance Services	Write audit-trail entries for every state-changing action